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Mr. McLane

ENGR 1304

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Compact Duct Exhaust Outlet

For this project, I designed an adapter that takes a 4-inch fan, and vents it out via a rectangular output with flaps to keep debris from going back through when air is not flowing. The assembly begins with the fan duct casing, just so that the end user can see what the adapter connects to, and that connects to the adapter. The adapter is made up of two primary parts. It has the transition piece, which takes the circular duct input and converts it to a rectangular output, and it has a piece that mounts to that which holds the flaps and mounting geometries. However, there are many more parts that went into this design/assembly.

The joints used are found at the hinges for the flaps, and on top of the flap casing, where a latch is found. To connect the flaps to the flap casing, they are press fit. To achieve this, the flap casing is made up of TPU (Thermoplastic Polyurethane), which has more flexible properties compared to other plastics, such as ABS (Acrylonitrile Butadiene Styrene), used for the other pieces. Because of the way that the joining mechanism works, the flaps are easily able to slide in given the flexibility of TPU, however, due to the shape of the joinery, they are unable to come out during typical use. Another example of this press-fit joinery is found on the top latch. The top latch connects at the top and serves as a way to mount the device to any fixtures necessary. Once again, because the casing is flexible, the latch is able to press fit on, and stay on during normal usage. Alternative ideas to this mounting were considered, such as mounting bearings and making the whole assembly two pieces that could be screwed together, however, the press-fit joinery was decided upon due to its simplicity and functionality. Every single hinge found on this design is a 1 DOF (Degree of Freedom) joint, which keeps things simple and secure.

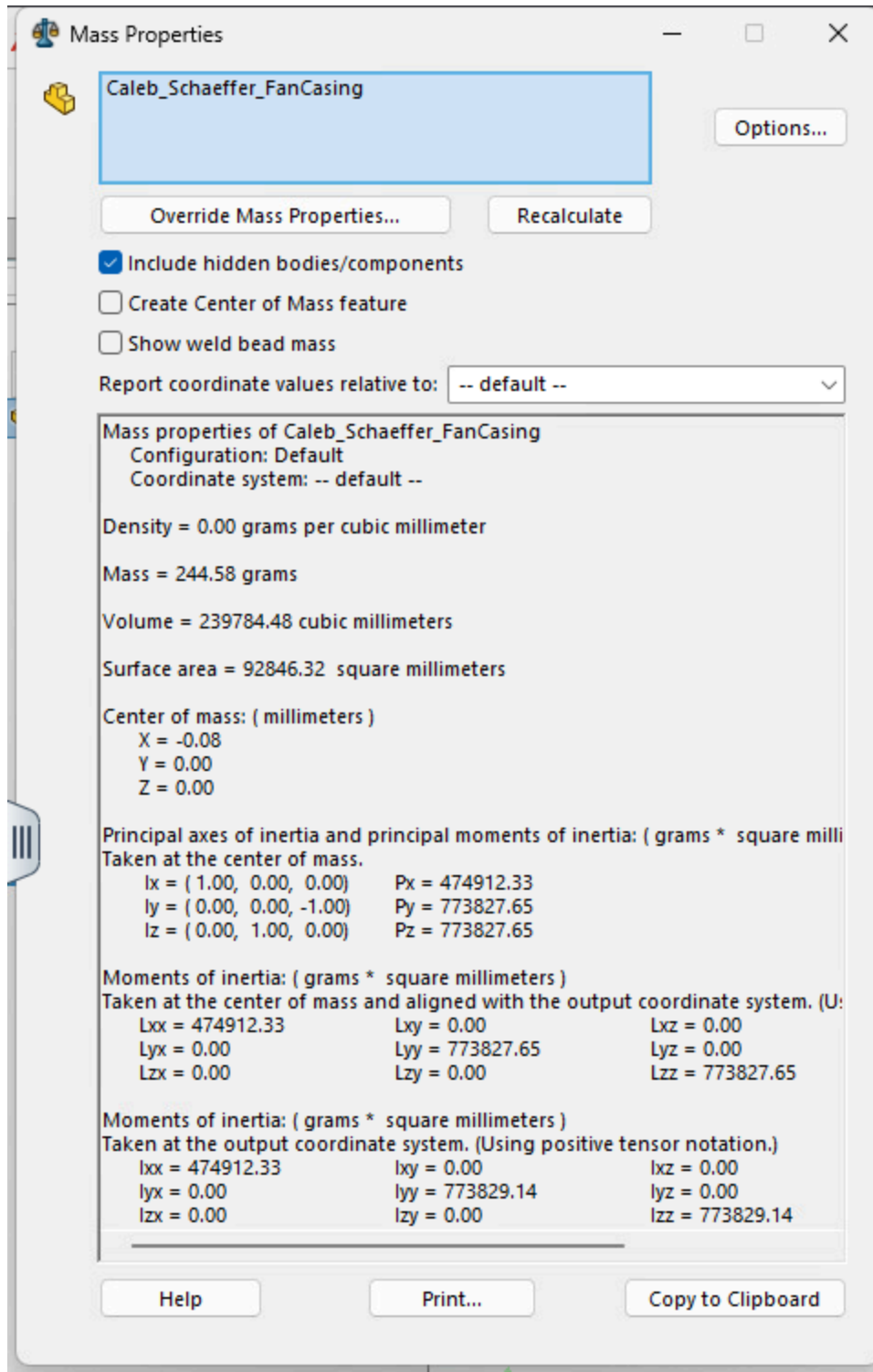
There were many design considerations that were taken into account for this project to come to life. One of them was the understanding that additive manufacturing was to be used for the final manufacturing of the device. As such, the materials are made in additive manufacturing friendly materials, such as ABS and TPU. The only pieces that are metal are standardized screws, such as the M2 screws. As such, the entire assembly is based on the metric system due to my personal familiarity with it as well as the dominance of metric in the additive manufacturing

space. The design had 3D printing as its preferred method of additive manufacturing at heart, and as such many more design considerations were taken for that. The majority of the design is able to be printed without supports, and the walls are all multiples of 0.4mm, with most of them being 4mm thick. This is important, as 0.4mm is the standardized nozzle size for most 3D printers, and most walls are able to be set to 4mm thick, meaning that you get clean surfaces and extrusion from a 3D printer. Another consideration that was made specifically in the spirit of 3D printing was the chamfers. Due to a 3D printing phenomenon called “elephant’s foot” parts may come out with the bottom surfaces that are attached to the bed looking bulged. Adding a chamfer of ~0.2mm helps with this, as if such a phenomenon happens, then it is quickly hidden away, and keeps the corner looking perfectly square. The design is also able to be modified as to the end user’s goals. If the end user is using a different duct fan, then they are able to edit the dimensions of the adapter to meet their parameters. Clearance was also important to the project due to the dimensional accuracy of the 3D printer used, however, in SolidWorks, these are not added in most sections. This is because most 3D printer users are familiar with their printers, know the shrinkage per material, and know the overall dimensional accuracy of their 3D printer. Due to this, the end user is able to go back into the SolidWorks file and edit the file to their specification. All of these considerations came together to make a design that works elegantly, is able to be manufactured easily, and can be easily scaled.

A few technical difficulties were found along the way for this design. One technical difficulty I faced was the usage of SolidWorks. Due to not being as familiar with SolidWorks as I am with other CAD programs, I found that there were particular functions or sketching tools that I needed that I was unable to find quickly. Eventually, I was able to find everything that was needed, however, this led to a significant time bump. Another technical difficulty was with poor planning as to the design of the project. At the start of the project, I did not have the dimensions for everything, and I was not certain what the end result would fully look like. To minimize any issues and further complicating things, I made the majority of the project in another CAD program that I am more familiar with to make sure everything worked up to spec, and then moved it over to SolidWorks. However, the majority of the designing was still done in SolidWorks, until I wasn’t certain how I wanted a certain piece to turn out as.

Standard M2 screws were not designed by me. They were created by Ioan Skurka and posted on GrabCAD.

Material Properties:



Mass Properties

Caleb_Schaeffer_AdapterPiece

Options...

Override Mass Properties... Recalculate

☒ Include hidden bodies/components

☐ Create Center of Mass feature

☐ Show weld bead mass

Report coordinate values relative to: -- default --

Mass properties of Caleb_Schaeffer_AdapterPiece
 Configuration: Default
 Coordinate system: -- default --

Density = 0.00 grams per cubic millimeter

Mass = 261.51 grams

Volume = 256379.64 cubic millimeters

Surface area = 98721.99 square millimeters

Center of mass: (millimeters)
 X = 0.00
 Y = 0.00
 Z = -34.55

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)
 Taken at the center of mass.
 Ix = (0.00, 0.00, 1.00) Px = 675303.76
 Iy = (1.00, 0.00, 0.00) Py = 693926.80
 Iz = (0.00, 1.00, 0.00) Pz = 866030.99

Moments of inertia: (grams * square millimeters)
 Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)
 Lxx = 693926.80 Lxy = -6.60 Lxz = 0.00
 Lyx = -6.60 Lyy = 866030.99 Lyz = -0.01
 Lzx = 0.00 Lzy = -0.01 Lzz = 675303.76

Moments of inertia: (grams * square millimeters)
 Taken at the output coordinate system. (Using positive tensor notation.)
 Ixx = 1006041.92 Ixy = -6.60 Ixz = 0.00
 Iyx = -6.60 Iyy = 1178146.11 Iyz = -0.01
 Izx = 0.00 Izy = -0.01 Izz = 675303.76

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